

Term: 2026 Summer Term (2 weeks course, 5 – 14 August)
 Class Schedule: 18:50 – 21:15 (with two 5-min breaks)
 Instructor: Professor Chin An Tan, Ph.D., Fellow ASME
 Dept. of Mechanical Engineering, Wayne State University, Detroit, MI 48202, USA
 email (tan@wayne.edu), DingTalk or WeChat
 Contacts: email (tan@wayne.edu), DingTalk or WeChat
 Office Hours: About 30 minutes before class in the classroom or by emails/texts

Course materials will be based on lecture notes developed by the instructor. The following is list of reference textbooks for further reading. Students are strongly encouraged to discuss with the instructor on any course-related matters.

Reference textbooks:

-  Rao, S. S., *Mechanical Vibrations*, 2018, 6th Edition, Pearson. ISBN-13: 978-1-292-17860-8.
-  Inman, D. J., *Engineering Vibration*, Prentice Hall, Fourth edition, 2014, ISBN-13: 978-0132281737.
-  Meirovitch, L., *Fundamentals of Vibrations*, Waveland Press, 2001, ISBN 1-57766-691-7.
(textbook; focus more on the fundamental theory and mathematics)
-  Ginsberg, J. H., *Mechanical and Structural Vibrations: Theory and Applications*, John Wiley and Sons, 2001, ISBN 0-471-37084-3.
-  Palm, William J., III, *Introduction to MATLAB for Engineers*, McGraw-Hill, Third Edition, 2011, ISBN 0-07-353487-0. (Reference for computational problems)

Course Objectives:

The goals of this course are to equip students with a fundamental understanding of introductory concepts in vibration with engineering applications and to develop skills for research and practice in vibration engineering. Emphasis is put on bridging theory to practice and the development of skills which include modeling and analysis, and numerical solution by MATLAB/Simulink.

Homework: Homework assignments will be assigned after each lecture or every other lecture, depending on the progress of the class. No late assignment will be accepted, except under unforeseen circumstances. **Submit your homework in class (in printed version) by the due date.**

Tests: Final exam, Few quizzes (roughly every 1 or 2 lectures)

Grading:	Homework	25%
	Project	20%
	Quizzes & in-class tests	25%
	Final Exam	25%
	Class attendance	5%
	Active classroom participation	bonus

Course final grades will be based on percentages scaled with respect to the highest total score or 950 points (maximum 1,000 points), whichever is higher.

Guidelines for assigning grades:

(A = 95%+, A- = 90%+, B+ = 87%+, B = 83%+, B- = 80%+, C+ = 77%+, C = 73%+, C- = 70%+, D+ = 65%+, D = 60%+, D- = 55%+, F = less than 55%)

Useful web links:

Modal Analysis Group on LinkedIn

Professor Dan Russell animation website <http://www.acs.psu.edu/drussell/demos.html>

LECTURE	OUTLINE OF TOPICS
<i>Foundation and Basic Concepts of Vibration of SDOF Systems</i>	
August 05 (Wednesday)	Course administration Introduction to vibration; Free vibration of single-degree-of-freedom undamped and damped systems; modeling and equations of motion; concept of natural frequency; response solutions; Measurement of damping; examples and MATLAB solutions
August 06 (Thursday)	Forced vibration of SDOF systems to harmonic excitations Frequency response solutions; concept of resonance, examples and MATLAB solutions
August 07 (Friday)	Vibration isolation techniques; Forced response of SDOF systems to arbitrary non-periodic excitations; convolution integral, Laplace transformation method, other solution methods; examples and MATLAB solutions
<i>Vibration of 2DOF Systems</i>	
August 10 (Monday)	Free vibration of two-degrees-of-freedom systems; modeling and equations of motion, Lagrange's equations Concepts of coupling, decoupled equations of motion and modal analysis
August 11 (Tuesday)	Forced vibration of 2DOF systems Frequency response, examples and MATLAB solutions
<i>Vibration of Multi-Degree-of-Freedom Systems</i>	
August 12 (Wednesday)	Modeling of MDOF systems; equations of motion; Lagrange's equations Orthogonality of vibration modes (modal vectors); modal analysis; necessary and sufficient conditions for modal decoupling and physical significance Examples and MATLAB solutions
August 13 (Thursday)	Free and forced responses of undamped systems Proportionally damped systems; examples and MATLAB solutions Discussion of Project
August 14 (Friday)	Free and forced responses of damped systems Discussion of Project Introduction to modal analysis (optional)
August 17 (Monday)	Final Examination
August 19 (Wednesday)	Project due at 23:00